AegisLife Insurance

# Risk Analytics: Executive Report

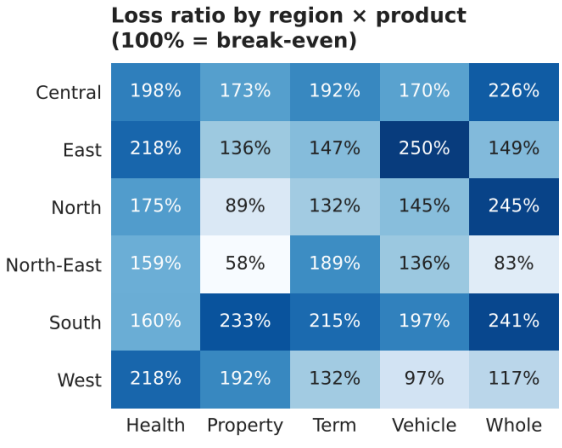
Prepared for the executive team | Data: 1,648 customers, 2,828 policies, 1,406 claims, 300 agents, 1,000 feedback records

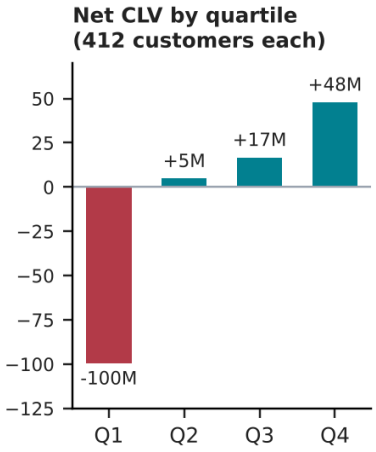
**Bottom line:** approved claims are 127.4M against 75.5M of annual premium, a loss ratio of 169%. For every 100 collected, 169 is paid out. The problem is concentrated and fixable: a quarter of customers cause 98% of claims, price ignores risk, and part of the leakage is plain control failure.

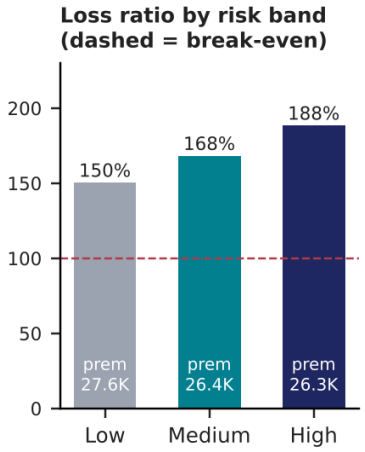
## Top 5 data-driven insights

1. **Almost every product and region loses money.** Only 4 of 30 region × product cells are below 100% (North/Property 89%, North-East/Property 58%, North-East/Whole 83%, West/Vehicle 97%). Worst: East/Vehicle 250%, South/Property 233%, Central/Whole 226%.
2. **A small group drives the loss.** The lowest-CLV quartile (412 customers) generates 98% of approved claims: net CLV of −100M, while the top quartile adds +48M.
3. **Price does not follow risk.** Average premium is almost the same across Low, Medium and High risk (27.6K, 26.4K, 26.3K), but the loss ratio climbs from 150% to 188%. Smokers average 134K per claim against 117.5K.
4. **Claims leakage and a weak fraud signal.** 89 approved claims exceeded policy cover (12.4M over). 49% of claims are flagged for fraud, and 248 flagged claims (61.5M) were still approved. 24% of claims take more than 45 days.
5. **Service and retention are weak where customers feel it.** Feedback on claim delay (1.9/5), documentation (2.0), pricing (2.1) and coverage clarity (2.2) is poor, against about 4.5 for agents and general service. The 10 highest-lapse agents lose 29% of policies versus a 14.9% average.

## Supporting visuals (Python analysis; same figures as the Power BI dashboard)




*Loss ratio = approved claim amount ÷ annual premium. CLV = premium earned − approved claims. Currency is not stated in the data.*

## Strategic recommendations, with evidence and impact

| Recommendation | Evidence | Estimated impact (stand-alone) |
|---|---|---|
| **1. Cap approved payouts at the policy cover amount; add an automatic system check** | 89 approved claims paid above cover, 12.4M extra (266 claims exceed cover) | −16 pts loss ratio (169% → ~152%) at almost no cost |
| **2. Fraud-investigation focus: review the 248 approved claims that carry a fraud flag** | 49% of all claims are flagged, yet 248 flagged claims worth 61.5M were approved; the flag does not separate good from bad claims | Each 10% of that value recovered = −8 pts; also re-tune the flag so investigators see fewer, better cases |
| **3. Risk-based repricing for Medium/High risk and smokers** | Premium is flat (26.3–27.6K) while loss ratio rises 150% → 188%; smokers claim 134K vs 117.5K | +10% premium = −15 pts (169% → ~153%); targeted at High risk gives more per customer lost |
| **4. Re-underwrite or reprice the 100 worst customers** | 100 customers: 57.9M claims (77 pts of loss ratio) vs 5.4M premium | Without them the book runs near break-even (~99%) |
| **5. Retention and claims-speed programme; targeted training for the weakest agents** | Top-10 agents lapse 29% vs 14.9% average; 24% of claims take > 45 days; claim-delay feedback scores 1.9/5 | Target: 45-day SLA on 90% of claims, lapse of top-10 agents to the average; protects the profitable Q3–Q4 customers (+65M CLV) |

**Combined effect:** recommendations 1 + 3 (10% repricing) + 10% recovery on 2 take the loss ratio from 169% to about 130%; recommendation 4 is the route to break-even. Estimates overlap slightly and are scenarios, not forecasts.

**Data notes:** the data looks synthetic. Fraud and churn models perform near chance, so the flag should be rebuilt before it drives decisions. Lapse is 65.7% by policy status but about 15% in the agent table; the definitions must be aligned. Next step: pilot 1 and 3 on the next renewal cycle and track loss ratio monthly in Power BI.